Ex8: Interactive Dashboards with Tableau

Learning Objective:
The objective of this experiment is to enable students to understand and implement interactive data visualisation and dashboard design using Tableau.

Aim: To create an interactive Sales and Profit dashboard in Tableau using the Superstore dataset.

Problem Statement:
Create an Interactive Dashboard in Tableau using the Sample – Superstore dataset from kaggle that includes:

- Bar chart (Sales by Category or Region)
- Line chart (Sales trend over time)
- Scatter plot (Profit vs Sales by Sub-Category)
- Interactive filters and dashboard actions

Tasks to Perform:

1. Open Tableau and Load dataset.
2. Explore the Data
3. **Create Worksheet 1 – Bar Chart (Sales by Category)**
   a. Click the **Sheet 1** tab → rename it `Bar_Sales_by_Category`.
   b. Drag `Category` to **Columns**.
   c. Drag `SUM(Sales)` to **Rows**.
   d. Sort the bars in descending order.
   e. Drag `Region` to **Color** → this gives different colors for each region.
   f. Add **Labels**: Click **Label** → check "Show Mark Labels".
   g. (Optional) Add filter: Drag `Region` to **Filters** → select all → then right-click it → "Show Filter".
4. **Create Worksheet 2 – Line Chart (Sales Trend Over Time)**
   a. Click **New Worksheet** → rename it `Line_Sales_Trend`.
   b. Drag `Order Date` to **Columns**.
   c. Right-click `Order Date` → select **Month** (continuous).
   d. Drag `SUM(Sales)` to **Rows**.
   e. Drag `Region` to **Color** (to show separate lines per region).
   f. Format the axis (if needed) → Right-click axis → "Format" → Currency.
   g. Show filter for Region or Category (optional).
5. **Create Worksheet 3 – Scatter Plot (Profit vs Sales)**
   a. Click **New Worksheet** → rename it `Scatter_Profit_vs_Sales`.
   b. Drag `SUM(Sales)` to **Columns**.
   c. Drag `SUM(Profit)` to **Rows**.
   d. Drag `Sub-Category` to **Detail** (each point = one Sub-Category).
   e. Drag `Category` to **Color**.

f. Optionally:
    i. Drag `Segment` to **Shape** (to add another dimension).
    ii. Add **Reference Line**: Analytics Pane → drag "Average Line" to both axes → "Table".

6. **Build the Dashboard**
    a. Click **New Dashboard** (bottom tab) → rename it `Sales Dashboard`.
    b. On the left, in the **Sheets** pane, drag and drop your 3 worksheets:
        i. Line chart at top
        ii. Bar chart bottom-left
        iii. Scatter plot bottom-right
    c. Adjust size (e.g., 1200×800).
    d. Use **containers** (horizontal/vertical) for cleaner alignment.
    e. Add **Title** → "Sales and Profit Dashboard".
7. **Add Interactivity**

(a) Add Filters

1. In Dashboard view → click dropdown on top-right of each sheet → "Show Filter".
2. Show filters for fields like **Region**, **Category**, or **Year**.

(b) Add Actions

1. Menu → **Dashboard → Actions → Add Action → Filter…**
2. Example:
    - **Source Sheet**: Bar chart (Sales by Category)
    - **Target Sheets**: Line and Scatter plots
    - **Run action on**: Select
    - **Clearing the selection**: Show all values
3. Add another **Highlight Action** (optional):
    - So hovering over a Sub-Category highlights the same category in other charts.

8. **Publish or Export**
    a. **To Tableau Public**: *File → Save to Tableau Public → sign in → publish.*
    b. **To PDF/Image**: *File → Export → Image/PDF.*

Solution:

The bar chart, line chart and scatter plot were created in separate worksheets by following the steps above and combined in the dashboard "Sales and Profit Dashboard". Region and Category filters and a filter action from the bar chart to the other two charts were added.

Result:

An interactive dashboard with a bar chart, line chart, scatter plot, filters and actions was created in Tableau.